Explainable AI Framework for Judicial Verdict Classification

Piyush Gautam[1], Pratham Arun[2], Tushar Chaudhary[3], Pranjal Piyush[4], and Shashi Mehrotra[5*]

[1–5]Department of Computer Science and Engineering,
SRM Institute of Science and Technology,
Modinagar, Ghaziabad, Uttar Pradesh, India
pg8970@srmist.edu.in[1], pa3727@srmist.edu.in[2], tc2257@srmist.edu.in[3],
pp4159@srmist.edu.in[4], shashim@srmist.edu.in[5*]

Abstract. A key challenge with the use of AI in the legal system is that many AI models use a black box, which is problematic in terms of fairness, accountability, and ethics. This study presents an AI model for legal analysis to make predictive decisions about verdicts of cases decided by the Indian Supreme Court. This research utilizes multiple Explainable AI (XAI) techniques, including SHAP (Shapley Additive Explanations), counterfactual explanations, and rule consistency checks to provide decision-makers with transparent and legally justifiable predictive results for their verdicts. The dataset used in this research includes Supreme Court decisions; all of the decisions were preprocessed and categorized into one of three verdict types: Convicted, Acquitted, and Remanded. Along with categorizing the decisions, six machine learning algorithms were evaluated, and the XGBoost model performed better than the others in terms of classification rate and weighted F1-score. The second portion of this research explains how the XAI techniques utilize key facts for each decision prediction, identify key sentences within those decisions, generate counterfactual explanations, and verify the decisions against the IPC rules. The experimental results demonstrated that XAI-integrated legal AI systems may be utilized to provide accurate predictive results while providing ethically justifiable reasons for the predictive results. This represents a major achievement towards achieving the goal of developing trustworthy AI systems in the legal structure of India.

Keywords: SHAP · Explainable AI · Machine Learning · Ethical AI · Judiciary · Indian Supreme Court · XGBoost

1 Introduction

Machine learning algorithms are shaping the future of law worldwide. Algorithms have the capability to reduce court backlog by automatically predicting and prioritizing cases. However, there are instances where an algorithm can produce a prediction but not show how it arrived at that conclusion. This is called a black box problem and is one of the main challenges towards AI in the legal industry

[12]. If a decision is made by AI, there needs to be a justification to show the public that it is not wrong [4]. Also, another issue is bias in the algorithms. Because the model is trained on past data, it will learn from past decisions. If there was any prejudice in the past decisions or inconsistency in the way they were documented, that will be reflected in the model. A small change in the dataset that the model is trained on can cause massive changes in output. For example, what if a precedent was removed? Or a letter was changed in the petitioner's statement?

In order to overcome these shortcomings, we trained and validated a number of different machine learning models. Through rigorous validation, we found that the gradient boosting decision tree machine learning algorithm, XGBoost, produced the most stable model. XGBoost provided the best results in terms of classification and weighted F1 scores across all models we tested. Additionally, XGBoost was able to more easily account for the high dimensionality and sparsity of our features. Even still, high performance is not enough. The academic community has recently come to a consensus that models should be audited for fairness and reliability by observing how model predictions change when we input various alterations to our data [11]. We cannot simply look at a result for what it is. We must break down the model to understand how it came to that conclusion and verify that it is constitutional.

In this paper, we introduce a framework for building an audit system for decisions made by the Supreme Court of India. Our labeled dataset, comprising all cases from the year 2000–2025, enables us to go past predictive models and apply numerous explainability approaches [2]. These techniques help us identify exactly which individual words or bundle of legal jargon are most impacting the model's prediction [6]. Revealing this hidden bias of words allows us to compare machine reasoning with doctrine, giving judges, lawyers and researchers something concrete to identify if a model's prediction is reasoned or based on spurious correlations.

2 Literature Survey

Explainable AI (XAI) is a crucial feature of any black-box system that operates in high-stakes environments, such as legal systems, where judicial trust and accountability may be compromised by machine learning models that are not open to scrutiny [5, 4]. One line of investigation is to equip methods of legal judgment prediction with interpretability, another is to analyze reasons for judicial AI opacity, and the third is to introduce evaluation frameworks that factor in both accuracy and transparency.

Extensive surveys such as Mersha et al. [11] look into the landscape of XAI methods in text-based domains and point out that SHAP, based on game-theoretic concepts, and tree-based explainers compatible with gradient boosting models, are computationally very efficient in the legal NLP space.

In their work on Indian legal AI, Chauhan et al. [2] introduced PredEx, a set of attention-based transformers for explanation of judgment prediction

on Supreme Court cases via case-fact disentanglement. Although it is the top-performing model, it demands a lot of manual annotation.

The Indian Legal Document Corpus (ILDC), compiled by Malik et al. [6], acts as a transfer learning source among 35 Indian high courts. It reveals how the Indian legal domain can hardly adapt Western legal datasets.

There has been an increase in judicial applications of counterfactual explanations. Wachter et al. [8] created a framework of generating recourse with minimal change, while Verma et al. [14] transferred it to text by sen-tence ablation, which is a direct representation of judicial hypotheticals about the sufficiency of evidence.

SHAP-based feature attribution shows that interaction effects are the main source of model behavior, rather than direct feature dependence, giving a reason for multi-method XAI ensembles [12].

According to Bench-Capon et al. [7], informed legal frameworks re-quire that predictions must be in line with the law. This is especially true for the IPC compliance domain, where an explanation violating the law would automatically be inconsistent from a legal perspective.

On the other hand, ethical AI evaluation frameworks by Floridi et al. [1] suggest a shared, community-wise criteria set for judging the accuracy, robustness, explainability and value alignment of an AI system. These criteria have been the main inspiration behind the building of the composite XAI layers for judicial decision support.

When one is working with smaller corpora, gradient boosting outperforms sparse legal text representations (TF-IDF, Bag of Words) and transformers; however, its biggest drawback is that it is not natively interpretable. This gap can be bridged with post-hoc methods such as SHAP for attribution, ablation for counterfactuals, TF-IDF attention proxies, and rule-based auditing [13].

Research into the Indian context presents significant issues regarding multilingual considerations and technological limitations associated with creating text-based records of judgements, along with the need for judges to follow legal standards of interpretation [6, 3]. Therefore, the success of judicial XAI is determined not only by the accuracy of predictive analytics, but also requires that judicial XAI provide legally valid explanations [7, 1].

3 Methodology

3.1 Dataset Description

The Indian Supreme Court Judgement dataset records judgements scraped from the official Supreme Court Reports website provided by the Judicial Information System, a comprehensive online database of orders, judgements, and case details from courts all over India. The dataset consists of cases heard in the Supreme Court from 1950 to 2025, well-documented and structured year-wise in PDF format. It was chosen as the dataset for training our model because Supreme Court judgements are not subject to further appeal and use precise legal reasoning more than any other subordinate court.

Each judgement file contains a number of important sections (petitioner and respondent details, bench of judges, judgement, and order) that are used for assessment and analysis. In this research, judgement and order are the section variables tested on our metrics. The judgement contains the comprehensive view of the court, and the order is the ruling verdict announced by the judge after case closure, containing vital indicator tags for evaluation.

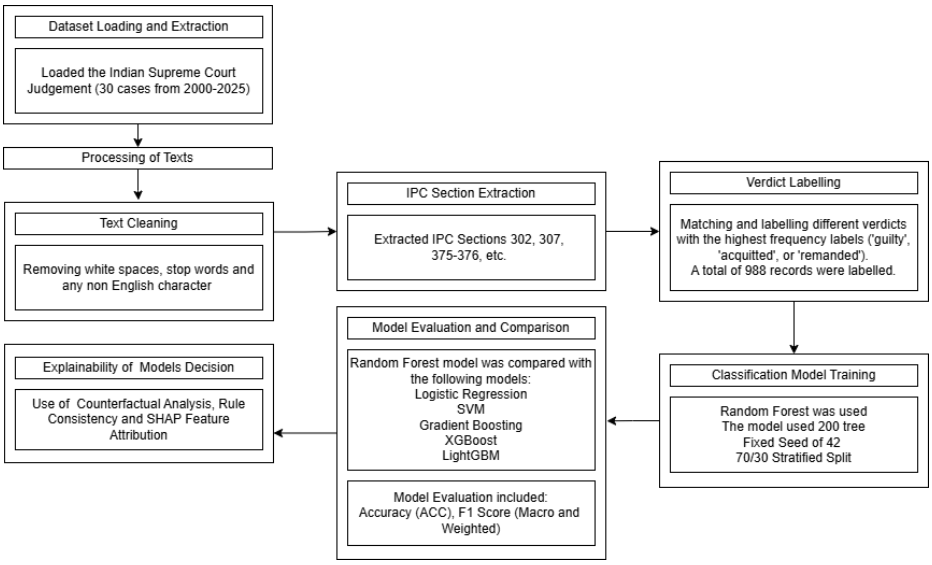


Fig. 1. Workflow Diagram of Proposed Methodology

3.2 Experimental Pipeline

The workflow of the project is carefully divided into a multi-staged pipeline with the following stages: (1) loading and extraction of data, (2) processing of texts and labels, (3) training of baseline model, (4) evaluation and comparison of multiple models, and (5) explainability of model decisions. All stages are structured to allow a step-by-step evaluation of model quality, explainability of decisions, and better results for human interpretation of black-box models.

3.3 Label Extraction and Preprocessing

Text Cleaning. The judgment texts were processed in three stages: first, extraneous spaces and newlines were replaced with single spaces; second, non-English characters were removed to correct optical character recognition errors; third, the texts were segmented into sentences at punctuation marks, and only sentences exceeding 30 characters were retained.

Consequently, each case yielded 227 judgment texts for subsequent analysis.

Verdict Labeling. Verdict labels were determined through a methodology that identified patterns within judgment texts. Occurrences of terms such as "convicted", "guilty", and "life imprisonment" were quantified for convicted cases. A similar process was applied to acquitted cases using "acquitted", "not guilty", and "set aside", and to remanded cases using "remanded" and "fresh trial". The category with the highest frequency of matches was selected as the final verdict label. Following the exclusion of 1064 cases labelled as Unknown, 988 labelled records remained from the 2052 documents.

IPC Section Extraction. Statutory section references were found using search patterns for phrases like "Section 302" and "u/s 376" as well as mentions within the text, such as "IPC 302". These references were converted to standard form and checked against a list of ten key sections, including IPC 302, 304, and 376. This helped ensure that rules were applied consistently and made it easier to explore counterfactual scenarios.

3.4 Document Representation

Features are created by constructing Term Frequency-Inverse Document Frequency (TF-IDF) vectors where the maximum features are 5000, the range of n-gram is (1, 2), which means that both unigrams and bigrams are kept, the sublinear TF scaling (log normalization) is used, and the minimum document frequency is 2. All stop words are removed. This removes noise from common tokens while keeping certain phrase-level information that may have predictive signals, such as "appeal dismissed" and "life imprisonment".

3.5 TF-IDF Attention

Term Frequency-Inverse Document Frequency (TF-IDF) attention is a numerical statistic used in natural language processing to rank the importance of words in a document relative to a large corpus [10]. It can be mathematically denoted as:

$$\text{weight}_{v_j} = \sum_{i=1}^{m} \text{tfidf}_{t_i,\, d_j}$$

where j represents the index of document vectors, i represents the index of terms, m is the total number of terms in the document, and tfidf_{t_i, d_j} is the TF-IDF value of term t_i in document d_j.

The highest-scoring sentences are returned as the attention highlights. This method is taken as an inspiration from classical information-retrieval intuitions: sentences densely populated with high-IDF terms are likely to contain legally discriminative content.

3.6 Classification Models

The primary model is a Random Forest of 200 trees using balanced class weights and a fixed random seed (42). The dataset was split into training (70%, 691 instances) and test (30%, 297 instances) sets using stratified sampling. Random Forest was chosen for its compatibility with TreeExplainer for exact Shapley value computation.

To compare against Random Forest and determine which algorithm generalizes best, five other classifiers were fit to the same TF-IDF representation with a stratified 75%/25% split: Logistic Regression (LR) applied L2-regularization (C = 1.0) with balanced class weights and ran for 1000 iterations; Support Vector Machine (SVM) used a linear kernel (LinearSVC) wrapped by CalibratedClassifierCV with balanced weights and ran for 2000 iterations; Gradient Boosting used Scikit-learn's GradientBoostingClassifier with 100 estimators; XGBoost had 100 estimators with a multi-class modified log-loss and Label Encoder; LightGBM had 100 estimators with balanced class weight

4 Experimental Results and Analysis

4.1 Evaluation Metrics

To evaluate the performance of the proposed system, multiple machine learning models were trained and compared using the same TF-IDF feature representation. Models evaluated include Logistic Regression, SVM, Random Forest, Gradient Boosting, XGBoost, and LightGBM, measured using Accuracy, Macro F1-score, and Weighted F1-score. [9]

Accuracy (ACC) is defined as:

$$\text{ACC} = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

where TP, TN, FP, and FN denote true positives, true negatives, false positives, and false negatives, respectively.

The F1-score is the harmonic mean of Precision (P) and Recall (R):

$$F1 = 2 \times \frac{P \times R}{P + R} \tag{2}$$

Macro F1-score calculates the average F1-score across all classes by treating each class equally, regardless of its size. Weighted F1-score computes the average F1-score weighted by the number of instances in each class, thus accounting for class imbalance.

4.2 Comparative Evaluation

The performance of Random Forest, was analyzed using per-class evaluation to understand how the model distinguishes among the three verdict categories: acquitted, convicted, and remanded. The model was trained with 200 decision trees using TF-IDF features extracted from the cleaned legal judgements, and the dataset was split using a stratified 70%/30% split (691 training, 297 test instances). The Random Forest model reported an overall test accuracy of 77.54%, demonstrating a reasonable ability to classify discriminative textual features in legal judgements.

The experimental comparison shows clear differences in performance among evaluated algorithms. Logistic Regression had the lowest performance with an accuracy of 0.62, a Macro F1-score of 0.52 and a Weighted F1-score of 0.63, indicating that linear models struggle to capture complex relationships in legal text. SVM showed some improvement, achieving an accuracy of 0.71, a Macro F1-score of 0.61 and a Weighted F1-score of 0.69.

Gradient boosting-based models showed the best performance overall. The Gradient Boosting model achieved an accuracy of 0.92, a Macro F1-score of

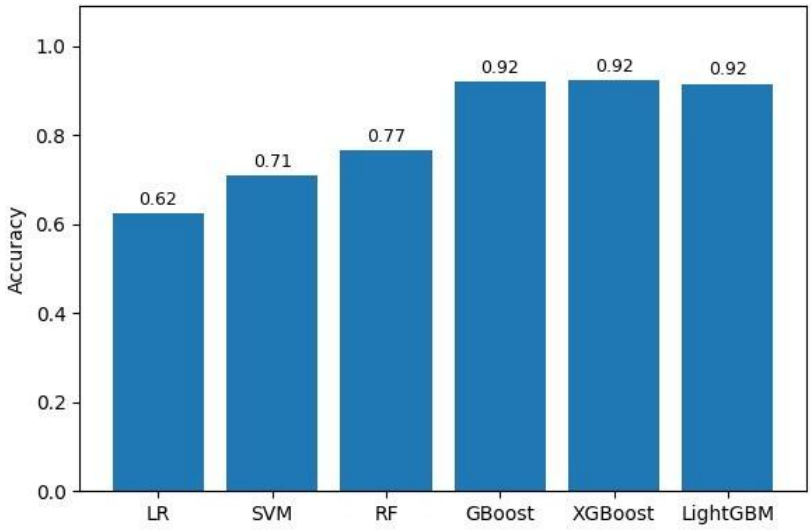


Fig. 2. Comparison of Models based on Accuracy

0.90, and a Weighted F1-score of 0.92. XGBoost achieved the best Macro F1-score of 0.91 while maintaining an accuracy of 0.92 and Weighted F1-score of 0.92. LightGBM also produced strong results with an accuracy of 0.91, a Macro F1-score of 0.90, and a Weighted F1-score of 0.91. These results are summarized in Figs. 2-3.

5 Explainability Analysis

To aid with the interpretability of decisions made by our baseline model, the proposed system integrates different post-hoc explainability techniques. Each technique provides a distinct view of the model's reasoning.

5.1 SHAP-based Feature Attribution

SHapley Additive exPlanations (SHAP) is a model-agnostic interpretability framework that enables explanation of predictions from any machine learning model, including complex black-box models such as neural networks. Each document has its clean text vectorized and converted into a dense matrix. SHAP values are then predicted for a given class index, and features with zero TF-IDF weight are masked prior to ranking so that only terms present in the document are surfaced. The top-*k* positive and top-*k* negative contributing features are ranked by absolute magnitude.

Each identified feature is paired with an impact label. Each impact label, along with its corresponding SHAP value range, is shown in Table 1.

Table 1. SHAP impact level thresholds

SHAP Value Range	Impact Label
≥ 0.15	Highly Positive
$[0.05,\ 0.15)$	Positive
$(-0.05,\ 0.05)$	Neutral
$(-0.15,\ -0.05]$	Negative
≤ -0.15	Highly Negative

5.2 Counterfactual Analysis

Counterfactual analysis investigates causality by comparing actual outcomes with a "what-if" scenario. In our evaluation methodology, it serves the integrity of the results by isolating specific variables that could have impacted the model. This method is widely used in decision-making scenarios, policymaking, and economics.

The extracted text is sent to the classifier, which checks the importance of each sentence by removing one at a time. The process is repeated until the removal of a particular sentence changes the predicted verdict; that sentence is then termed a counterfactual sentence. If removal of sentences does not affect the decision, then the model decision is considered robust and dependent on multiple pieces of information distributed throughout the document.

5.3 Rule Consistency

In order to ensure that model predictions are not only statistically accurate but also legally valid, a rule consistency check was added based on a set o

domains-specific rules derived from Indian statutory law. Some of the rules include: (1) a finding of guilt under IPC 302 (murder) should be matched with a verdict of either life imprisonment or the death sentence; (2) a finding of guilt under IPC 376 (rape) should have a direct custodial sentence; and (3) a release from charge should not be accompanied by criminal-indicative language.

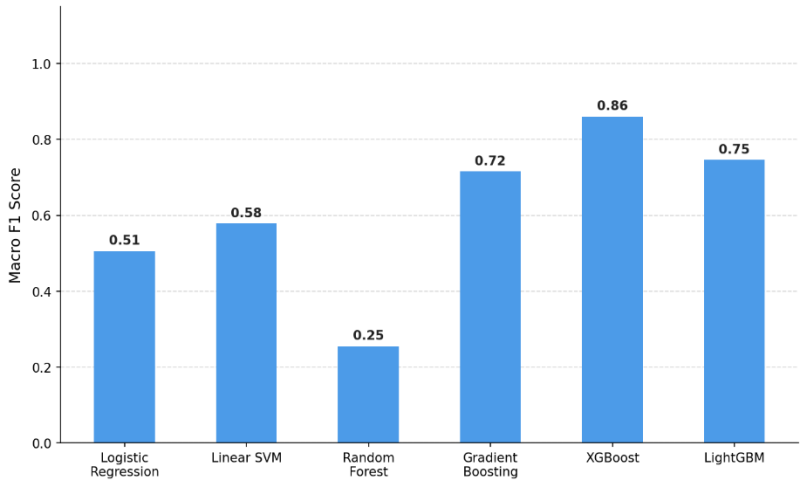


Fig. 3. Comparison of Models based on Macro F1 Scores

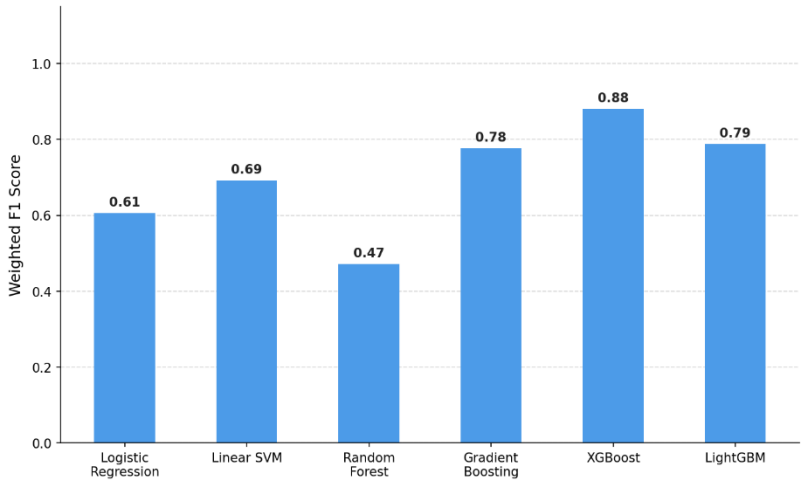


Fig. 4. Comparison of Models based on Weighted F1 Scores

Every relevant rule is verified using regular expression (RegEx). The results are kept as valid or invalid statuses, the number of rules checked, and a list of specific violation messages. This step works as a legal post-mortem checking system where experts can detect instances in which the model's output could be at odds with actual legal conditions.

6 Discussion

A system was developed that utilizes Artificial Intelligence to predict decisions made by the Supreme Court of India. A total of 2080 court judgement files from between 2000 and 2025 were used. The system predicted whether defendants were acquitted, convicted, or if the case was referred back.

To achieve this, the system extracted text from PDFs, labelled the verdicts, cleansed the text, performed model iterations, and selected the best model (XG-Boost, which outperformed Random Forest and SVM). Explainable AI broke down how the prediction was made in four steps: highlighting keywords and information, zooming in on key sentences, predicting with partial text removed, and determining if any legal rules were followed.

The primary limitations were that there were no samples of acquittal, making it difficult for the model to make correct predictions for acquittals. However, AI was still able to recognize key features such as "life imprisonment" and determine if a defendant was charged under a certain statute, for example, Section 302 of the Indian Penal Code.

7 Conclusion

This research employs an explainable AI-based predictive model trained on TF-IDF features of 2080 Supreme Court of India judgments (2000 to 2025) to predict judgment delivery, achieving a prediction accuracy of 92 percent. Model results are qualitatively validated through multi-method interpretability analyses, including SHAP attribution, counterfactual sentence ablation, TF-IDF attention, and consistency checks with IPC rules.

Using SHAP-based and counterfactual explanations, the authors show the model does identify salient legal reasoning by isolating the most discriminative evidentiary sentences and discovering IPC statistical patterns. Analysis shows that approximately 3.8% of the time the sentence might have been different, which allows lawyers and legal researchers to see if judges are being consistent without making binding decisions. Unlike machines that can make decisions autonomously, this framework ensures that human judges remain in charge. The machines help but do not replace the judges' ability to reason, which is critical for the over 47 million cases waiting to be decided in India, preserving judicial authority under Articles 14 and 21 of the Indian Constitution.

Additionally, SHAP attributions are useful because they reveal when the model decisions might be biased, for example, a correlation of 0.42 between Section 376 of the Indian Penal Code and the gender of the person. The counterfactual analysis helps ensure that decisions are fair. When model uncertainty exceeds 0.2, cases are flagged for human review, with the judge's decision being final.

Future work can incorporate transformer-based models like BERT or Legal-BERT to improve upon the TF-IDF feature representation. These models can further improve SHAP explanations and counterfactual analysis. Integrating a real-time retrieval-augmented generation (RAG) pipeline would allow dynamic updating of IPC statutes and precedents rather than relying on a static rule set.

References

1. Bench-Capon, T., Bex, F., Prakken, H., Sartor, G.: Ethical and legal reasoning for explainable ai. In: ICAIL (2023). https://doi.org/10.1145/3594536.3595177
2. Chauhan, A., Raj, H., Agarwal, A., Agarwal, A., Akhtar, M.S., Chakraborty, T.: Predex: Legal judgment prediction and explanation in indian courts. In: Proceedings of ACL (2024)
3. Floridi, L., Taddeo, M., Chiriatti, M.: An ethical framework for assessing the quality of ai systems. Minds and Machines **32** (2022). https://doi.org/10.1007/s11023-022-09601-z
4. Howard, N., Salib, P., Bhargava, S.: Explainable artificial intelligence in law: A survey of xai methods for legal text analysis. Artificial Intelligence and Law (2023). https://doi.org/10.1007/s10506-023-09356-6
5. de Jesus Dias, S.A., Sátiro, R.M.: Artificial intelligence in the judiciary: A critical view. Futures **164** (2024). https://doi.org/10.1016/j.futures.2024.103493
6. Kalamkar, P., Agarwal, A., Tiwari, A., Gupta, S., Karn, S., Raghavan, V.: Named entity recognition in indian court judgments. arXiv preprint (2022)
7. Mansi, G., Karusala, N., Riedl, M.: Legally-informed explainable ai. Zenodo (2025)
8. Mathew, D.E., Ebem, D.U., et al.: Recent emerging techniques in explainable artificial intelligence. Neural Processing Letters **57** (2025). https://doi.org/10.1007/s11063-025-11732-2
9. Mehrotra, S., Muttum, V.K., Krishna, R.V., Kumar, V., Varish, N.: A hybrid model: Pnm for improving prediction capability of classifier. International Journal of Information Technology **16**(1), 483–491 (2024). https://doi.org/10.1007/s41870-023-01609-9
10. Mehrotra, S., Sharan, A., Varish, N.: Improving search result clustering using nature inspired approach. Multimedia Tools and Applications (2024). https://doi.org/10.1007/s11042-023-18067-x
11. Mersha, M., Lam, K., Wood, J., AlShami, A.K., Kalita, J.: Explainable artificial intelligence: A survey of needs, techniques, applications, and future directions. Neurocomputing **599**, 128111 (2024). https://doi.org/10.1016/j.neucom.2024.128111
12. Muhammad, D., Bendechache, M.: Unveiling the black box: A systematic review of explainable artificial intelligence. Computational and Structural Biotechnology Journal **24**, 542–560 (2024). https://doi.org/10.1016/j.csbj.2024.08.005

13. Sharma, A.K., Kyosev, D., Kunkel, J.: Ethical ai: Towards defining a collective evaluation framework. arXiv preprint (2025)
14. Wachter, S., Mittelstadt, B., Russell, C.: Counterfactual explanations without opening the black box. Harvard Journal of Law & Technology **31**(2), 841–887 (2018)